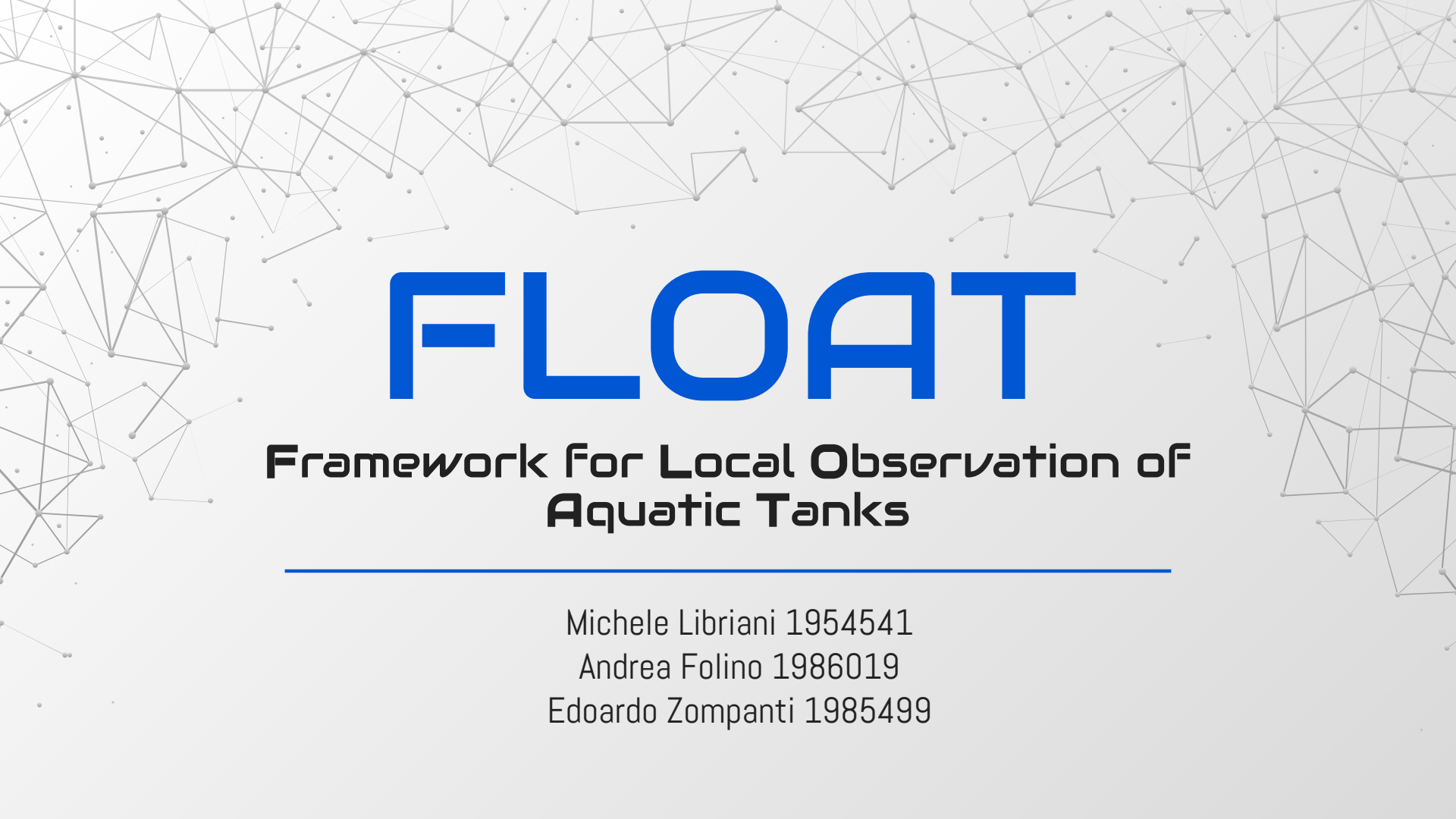




FLOAT

**Framework for Local Observation of
Aquatic Tanks**

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The Problem



Pump Failure

- Motor Damage
- Low Oxygenation



Water Quality

- High Turbidity
- Bacterial Blooms



Temperature

- Excursion
- Environmental shock

Existing Approaches

Approaches	Motor Monitoring	Offline Capable	Autonomous Protection
Manual checks	✗	✓	✗
Cloud Aquarium Systems	✓	✗	✓
Smart feeders	✗	✓	✗
FLOAT ✦	✓	✓	✓

①

②

3

4

5



Requirements



Motor Anomaly Cutoff

REQUIREMENT: Motor cutoff < 2000 ms

METRIC: Reaction Time (3 ticks)



Environmental Warning

REQUIREMENT: Alerts for unsafe temperature or turbidity

METRIC: Persistency tracking (3 ticks)



Offline Functioning

REQUIREMENT: Autonomous execution without WiFi

METRIC: Connectionless T/O communication

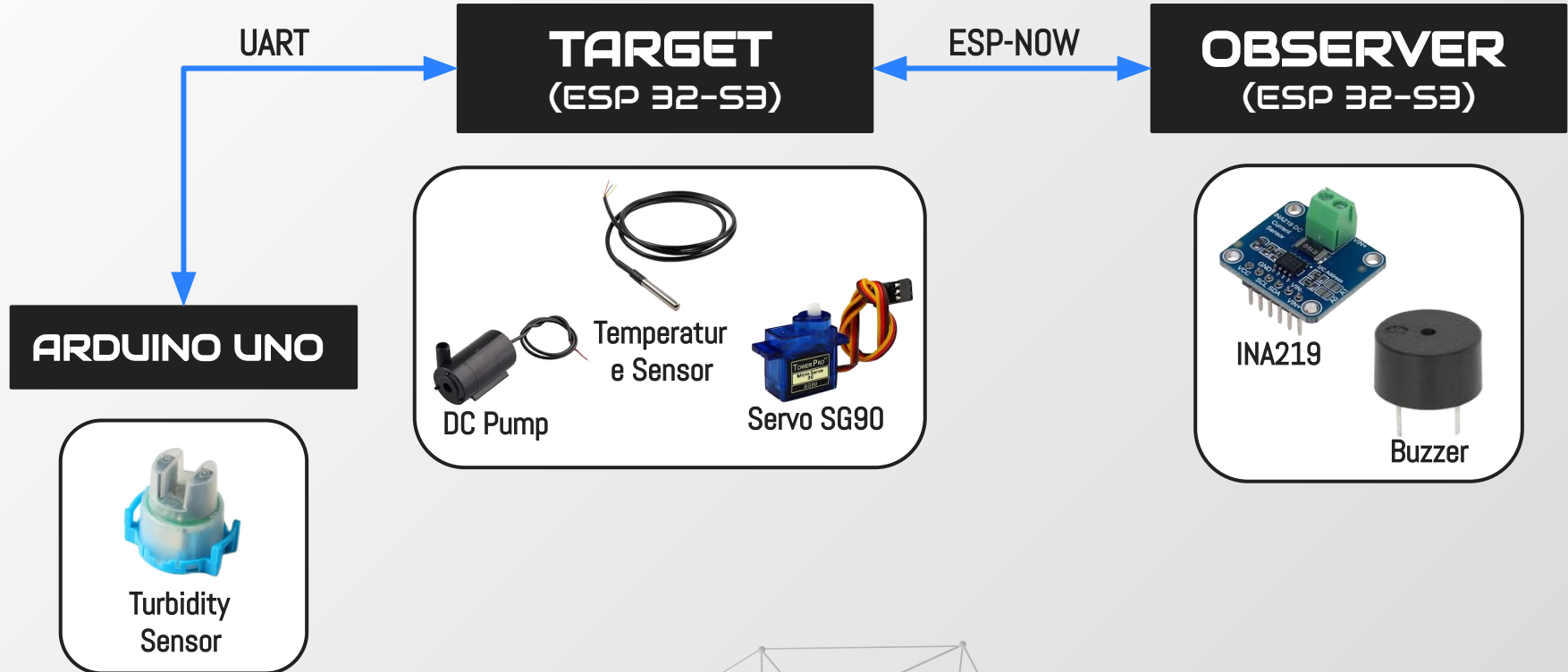


Detection Performance

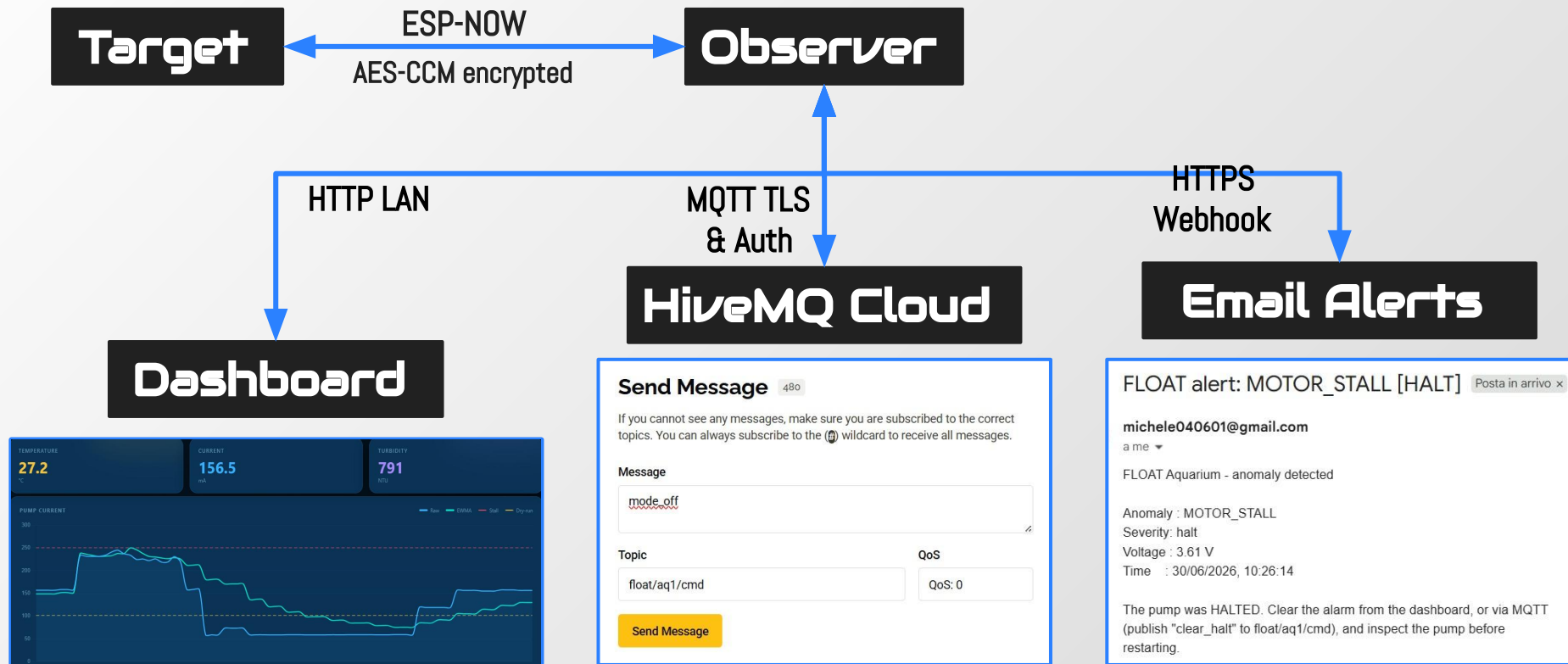
REQUIREMENT: Macro FPR/FNR $< 5\%$ & Accuracy $> 90\%$

METRIC: Confusion Matrix

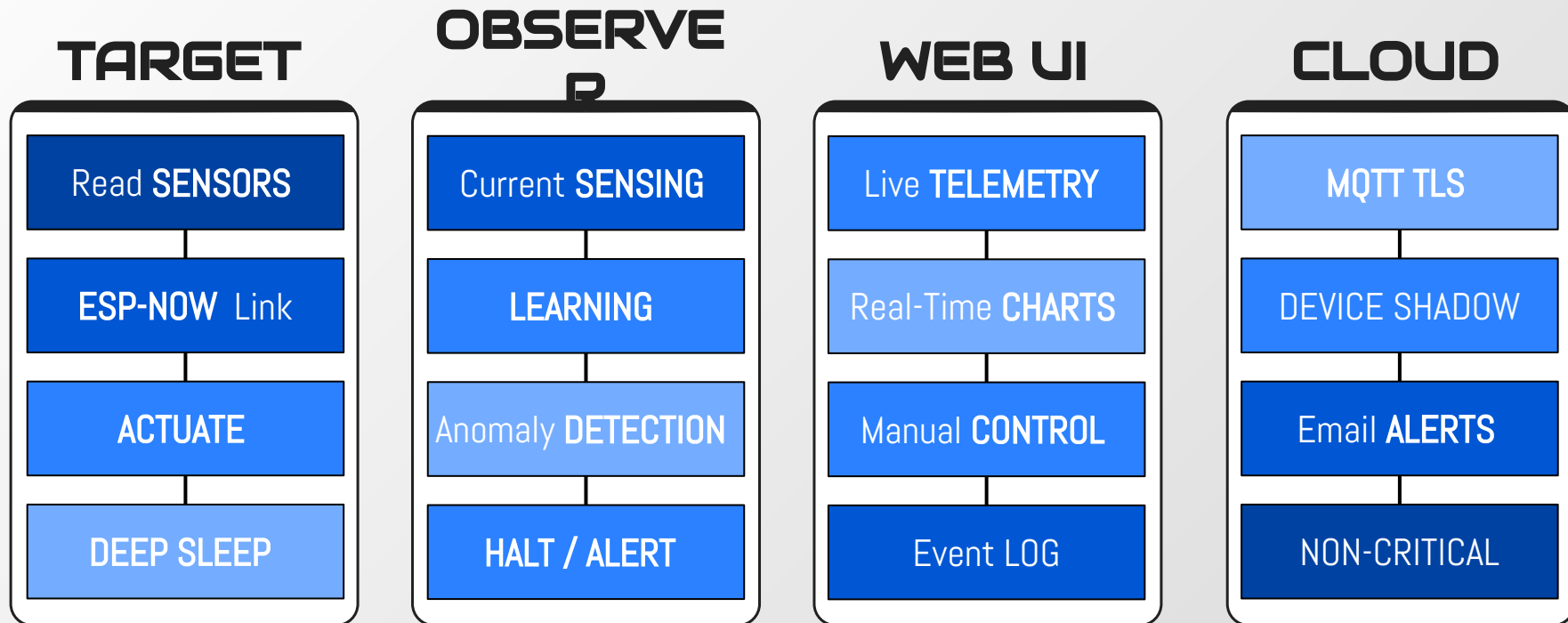
Hardware Components



Network Diagram & Security



Service Architecture



Motor Anomalies

MOTOR STALL

$EWMA > \max(\mu + 3\sigma, +15\text{mA})$

DRY RUN

$EWMA < \min(\mu - 3\sigma, -15\text{mA})$

① Calibration - Hampel Filter

$$MAD = \text{median}(|x_i - \text{median}(x)|)$$

$$\sigma_H = 1.4826 \times MAD$$

② Tracking & Mitigation - EWMA

$$ewma_t = \alpha \cdot I_{raw} + (1 - \alpha) \cdot ewma_{t-1} \quad \alpha = 0.2$$

③ Grace Period

4 ticks (1.6s) suppressed

④ Confirmation Gate

3 consecutive iterations $\rightarrow$ HALT

Advanced Diagnostics

VOLTAGE DROP

$$V < \max(3.3V, 90\% V_{\text{base}})$$

TEMP OUT OF RANGE

$$|T - T_{\text{learn}}| > \max(5 \sigma_T, 1.5^\circ \text{C})$$

LOW TURBIDITY

$$\text{Turbidity} < 100 \text{ NTU}$$

5 Predictive Maintenance - CUSUM

$$dev = (cycle_mean - \mu) - K \cdot \sigma \quad K = 1$$

$$cusum + = dev \quad (\text{clamped at } 0)$$

$$\text{IF } cusum > H \times \sigma \quad (H=3) \rightarrow \text{Alert}$$

6 Temperature Compensation

$$NTU_{\text{corrected}} = \frac{NTU_{\text{raw}}}{1 + 0.005 \times (T - 25)}$$

$$\text{Adaptive Temperature} \rightarrow \Delta = \max(5\sigma_T, 1.5^\circ \text{C})$$

System Evaluation

$$FPR_k = \frac{FP_k}{FP_k + TN_k} \quad \longrightarrow \quad FPR_{macro} = \frac{FPR_{normal} + FPR_{stall} + FPR_{dry}}{3}$$

$$FNR_k = \frac{FN_k}{FN_k + TP_k} \quad \longrightarrow \quad FNR_{macro} = \frac{FNR_{normal} + FNR_{stall} + FNR_{dry}}{3}$$

EVALUATION — CONFUSION MATRIX

Set the ground truth for the cycle you are about to run, then run monitored cycles (easiest in Default mode ON). Each cycle records one outcome.

Normal

Stall

Dry-run

Labelling off • samples: 43 • accuracy: 95.3% • macro FPR: 2.4% • macro FNR: 4.8%

truth \ det	NOR	STA	DRY
NOR	15	0	0
STA	1	14	0
DRY	1	0	12

Stop labelling

Reset matrix

Consumption Estimate

Components	Current
Heltec V3	~ 5-10 mA sleep ~ 40-50 mA awake
DC pump	~ 200-235 mA
Servo SG90	~ 35-45 mA avg
Esp-Now tx	~ 20-25 mA avg
Arduino UNO	~ 40-50 mA awake ~ 35-45 mA watchdog



Duty Cycle & Battery Life

PHASES

- Overhead Phase (3s) : Heltec Awake (45) + ESP-NOW (22) + Arduino (45) = **112mA**
- Servo Phase (2s) : Overhead (112) + Servo SG90 (38) = **150 mA**
- Pump Phase (5s): Overhead (112) + DC Pump (218) = **330 mA**
- Deep Sleep (20s): Heltec Sleep (7) + Arduino (45) = **52 mA**

SCENARIO

1. CLEAN WATER (PUMP 5s + FEED)

$$DutyCycle = \frac{3s + 2s + 5s}{30s} = \frac{10s}{30s} = 33.3\%$$

$$I_{avg} = \frac{(3s \cdot 112mA) + (2s \cdot 150mA) + (5s \cdot 330mA) + (20s \cdot 52mA)}{30s} \cong 111mA$$

$$Battery\ Life = \frac{5600mAh}{111mA} = 50.5h$$

2. DIRTY WATER (PUMP 10s)

$$DutyCycle = \frac{3s + 10s}{33s} = \frac{13s}{33s} = 39.4\%$$

$$I_{avg} = \frac{(3s \cdot 112mA) + (10s \cdot 330mA) + (20s \cdot 52mA)}{33s} \cong 142mA$$

$$Battery\ Life = \frac{5600mAh}{142mA} = 39.4h$$

Duty Cycle & Battery Life

DUTY CYCLE 2.5%

- Cycle (10min) : Duty Cycle = $(3s+2s+10s) / 600s = 2.5\%$
- Overhead Phase (3s) : Heltec Awake + ESP-NOW = **67mA**
- Servo Phase (2s) : Overhead + Servo SG90 = **105 mA**
- Pump Phase(10s) : Overhead + DC Pump = **285 mA**
- Deep Sleep (585s) : Heltec Sleep = **7 mA**
- $$I_{avg} = \frac{(3s \cdot 67mA) + (2s \cdot 105mA) + (10s \cdot 285mA) + (585s \cdot 7mA)}{600s} \cong 12mA$$

$$Battery\ Life = \frac{5600mAh}{12mA} = 467\ h \cong 19\ days$$

REAL AQUARIUM

- Cycle (24h) : Pump always ON $\rightarrow$ Duty Cycle = **100%**
- Pump Phase(43198s) : Heltec Awake + ESP-NOW + DC Pump = **285 mA**
- Servo Phase (2s) : Pump Phase + Servo SG90 = **323 mA**
- $$I_{avg} = \frac{(43198s \cdot 285mA) + (2s \cdot 323mA)}{43200s} \cong 285mA$$

$$Battery\ Life = \frac{5600mAh}{285mA} = 19.7h$$

Demo and Future Plans



Future Work

1. Remove Arduino Uno
2. LoraWan + Multi-Tank
3. Flash Encryption + Secure Boot
4. Advanced Water Chemistry
5. Active Heater Control